AAVISHKAR

Title of the research project

Integrated Railway Maintenance Prioritization and Constraint-Based Block Planning for Asset Availability

Category	Level	Slot / Code No.
5 — Engineering and Technology		

Problem

Track, traction, and signalling request blocks separately. Defects sit in TMS, SMMS, and TDMS. Requests go through BDMS. COA holds the corridor, and the working timetable is separate. [1] Mail/Express punctuality on Indian Railways was 73.62% in 2023–24 and 77.12% in 2024–25. Asset maintenance is one stated cause of delay. [2]

Research objective

Show one inspectable path from those records to a feasible possession. Urgency, the escalation forecast, and the block decision stay separate. A person still sanctions the window. Rolling Block, notified on 30 November 2023, is the operating idea this lines up with. [3]

Workflow · source records to a window a person can accept

TMS	SMMS	TDMS	COA	WTT
Track defects	Signalling	Traction	Corridor	Timetable

Intended feeds. This build uses file readers and an encoded 192 km CSMT–Pune corridor, not live servers.

1 Read and check

A bad row is rejected with a reason. The rest of the file still loads.

2 AI-1 review

Local rules propose a finding. The quote must appear on the page. A person applies ready rows.

3 Formula 1

Explainable urgency from severity, overdue days, a speed restriction, and traffic.

4 Days to critical

A separate forecast of days until severity 80. It does not change Formula 1.

5 Formula 2

Same line and feeder, next defect within 3 km, one possession.

6 CP-SAT

OR-Tools searches a legal window, or it reports that the window is infeasible.

7 Dashboard

Punctuality-First, Asset-First, and Mega-Block. A person sanctions, shifts, or rejects.

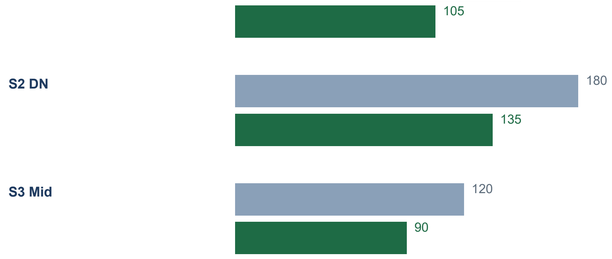
Severity, Formula 1, and the forecast are not added together. AI-1 does not rank the job and does not choose the block.

What each step adds

Formula 1 is the priority a controller can read: severity, overdue days, a speed restriction, and traffic. The forecast only estimates days until severity reaches 80, and its labels are synthetic. Formula 2 replaces repeated visits with the longest job plus 15 minutes. CP-SAT keeps a train and a gang apart, gives Premium and Superfast trains no induced delay, and caps the model fleet at 1 ballast cleaner, 2 tampers, and 2 tower wagons.

Encoded-backlog result

Job minutes. Not measured train delay.



Grey: each job alone. Green: one bundle. 450 minutes became 330.

Limits and next step

This is not a live deployment. The ordinary AI-1 reader matches known phrases and misses unfamiliar wording. The forecast has not been tested on real defect histories: about 1 day of error on a matched synthetic hold-out, and about 3.8 days when that synthetic process changes. Next evidence is authorised feeds, a shadow-mode trial, and a held-out set of real reports.

192 km	170	10 of 14	120 min
Encoded CSMT–Pune corridor	Services in the 24-hour model	Defects placed in one 7-day run	Saved on three clusters, 26.7%

References

[1] Ministry of Railways, SIH 2026 problem statement SIH26027. Separate TMS, SMMS, and TDMS records; BDMS requests; COA corridor availability.

[2] Lok Sabha Unstarred Question 3976, answered 12 August 2026. Mail/Express punctuality on Indian Railways: 73.62% (2023–24), 77.12% (2024–25), 77.43% (2025–26).

[3] Rajya Sabha Starred Question 62, answered 5 December 2025. Asset maintenance is one stated cause of delay. Rolling Block: Gazette notification 30 November 2023.

[4] CAG Report No. 45 of 2017, tabled 13 March 2018. Selected heavy-traffic sections. On 30 August 2017, Railway Board stated that non-availability of blocks was the main reason for overdue deep screening in those observations. Historical.

The four figures above were recomputed from the encoded demonstration backlog and the 24-hour timetable model. They are not field operating results. The 7-day run starts on 24 September 2026.